Subject

36103 - Statistical Thinking for Data Science

8 Credit points

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Subject description

This subject instills the statistical thinking central to data science, teaching you to apply statistical methods to extract insights from data. You'll start with fundamental principles and modern inference methods, gaining hands-on experience with real-world data. Progressing to statistical models and estimation techniques, you'll complete a data science research cycle, formulating questions, analyzing data, and communicating results effectively. The subject begins with workshops and lectures, transitioning to self-directed learning, fostering your growth as a data scientist. By the end, you'll have a solid foundation in the technical, conceptual, and practical realms of data science.

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Faculty
TD School

Subject type
Subject

Study level
Postgraduate

Result type
Grade and marks

Total Workload hours
200

Apply now

Important dates

Fee calculator

Availabilities

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Learning outcomes

Collapse all

Upon completion of this subject, graduates will be able to:

1. Manage data science project complexities.
2. Formulate precise data science inquiries and justify the use of appropriate statistical concepts and tools to respond to those questions.
3. Find, clean, and merge datasets from a range of sources to answer real world data science problems.
4. Apply appropriate statistical methods for datasets and interpret and communicate the results in ways that suit the needs of diverse stakeholders.
5. Engage with the local and global data science professional community to build ethical practices and responsibilities when working with Indigenous data and communities.

Learning and teaching activities

Authentic problem based learning: This subject relies heavily upon the principle that students learn best by doing. It offers a range of authentic data science problems to solve that help to develop students' statistical thinking about complex problems. Students work on real world data analysis problems using datasets that they create using modern data harvesting techniques. These are used to answer realistic data science questions in broad areas of topical interest. This exposes them to the true ambiguities, constraints, and complexities of working as a data scientist for a variety of different stakeholders.

Blend of online and face to face activities: This subject is offered through a series of block sessions blending online with face-to-face learning. Students interact face-to-face with each other and the teaching team in three intensive modules that require the completion of both preparation and after class activities. They concurrently use a range of complementary online resources to develop their statistical thinking according to identified weaknesses in their background knowledge. They are expected to engage in online discussion and to actively participate in other blended activities.

Collaborative work: We place a strong emphasis on group activities and collaboration in diverse teams. As a data science professional you need to approach professional projects and challenges by working with people from different backgrounds, expectations, and expertise. This course simulates that environment by requiring students to work with a team of peers who come from many different backgrounds. Group assessments help students to develop effective strategies for working as a part of a data science team, as well as an appreciation that there are diverse perspectives on many different topics in data science and innovation.

Self paced evaluation and improvement: This subject takes students from an exceptionally wide range of backgrounds, some of who are better versed in statistical methods, and Python, than others. We help all students to self-diagnose their weaknesses and strengths, and to work to improve in areas that they identify as a priority for the professional niche that they would like to occupy as a practicing data scientist. Students choose their own path through a wide variety of curated resources as needed.

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Assessment information

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Requisites

Requisite detail

Before you enrol in this subject, you must ensure you have met its requisites. The requisites for this subject can be viewed on its access conditions page.

To learn more about requisites, please visit the How to use the Handbook guide.

Additional subject details

Collapse all

Attendance requirement

block: on-campus; 6 x Monday evenings, on-campus; 1 x Saturday workshops, online; 2 x optional assessment support sessions, online; independent learning activities, online

UTS acknowledges the Gadigal people of the Eora Nation, the Boorooberongal people of the Dharug Nation, the Bidiagal people and the Gamaygal people, upon whose ancestral lands our university stands. We would also like to pay respect to the Elders both past and present, acknowledging them as the traditional custodians of knowledge for these lands.

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